

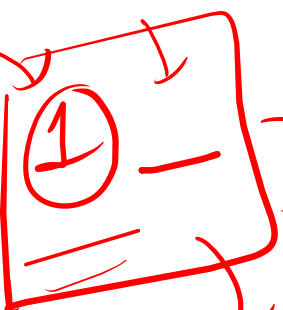
`print("111");`
↓
× 3

→ 1 1 1 ... 100
→ 1 1 1
→ 1 1 1

simplest way.

nested loop //

100 times

① →  100 times automated?

↓ ↓ ↘
`for(i=1; i<=100; i++) {`

↳ `for(j=1; j<=100;`

`j++)
 print(1 + " ");
} }`

$$n=4$$

$$i=1 \rightarrow 1$$

$$i=2 \rightarrow 1\ 2$$

$$i=3 \rightarrow 1\ 2\ 3$$

$$i=4 \rightarrow 1\ 2\ 3\ 4$$

$$i \rightarrow \begin{array}{c} 1 \dots \textcircled{i} \\ \underline{1\ 2\ 3} \dots \textcircled{i} \end{array}$$

code

input $n=4$

$$\rightarrow \begin{array}{c} 1 \\ 1\ 2 \\ 1\ 2\ 3 \\ 1\ 2\ 3\ 4 \end{array}$$

$$n=3$$

$$\rightarrow \begin{bmatrix} 1 \\ 1\ 2 \\ 1\ 2\ 3 \end{bmatrix}$$

line numbers → for(ⁿi=1; i<=n; i++) { n → changes

for(j=1; j<=i; j++) {

system.out.print(j + " ");

}

⊗ system.out.println(); ←

}

content

i=2: 2 // i-1

$n=3$ ✓

1

2 1

2 1

$i \rightarrow n-i$ spaces
 $i=1 \rightarrow$ reverse order //

$n=4$

✓	✓	✓	1
		2	1
	3	2	1
4	3	2	1

$i=1 \rightarrow 3$ spaces $\rightarrow 1$

$i=2 \rightarrow _ _ 2 1$

$i=3 \rightarrow _ 3 2 1$

$i=4 \rightarrow 4 3 2 1$

$i \rightarrow n-i$ spaces
 $i \rightarrow 1 \dots i$
 $i-1 \dots 1$

$n=4$

1			1				
2		1	2	1			
3	1	2	3	2	1		
4	1	2	3	4	3	2	1

$3 \rightarrow$
 1 space ✓
 1 2 3 $\rightarrow$ 1 $\dots$ i
 2 1 $\rightarrow$ i-1 $\dots$ 1

1 $\rightarrow$ 3 spaces 1 ✓
 2 $\rightarrow$ 2 spaces
 1 2
 1 i
 1 i-1 $\dots$ 1

$n = 4$
 $i = 3 \rightarrow$

①

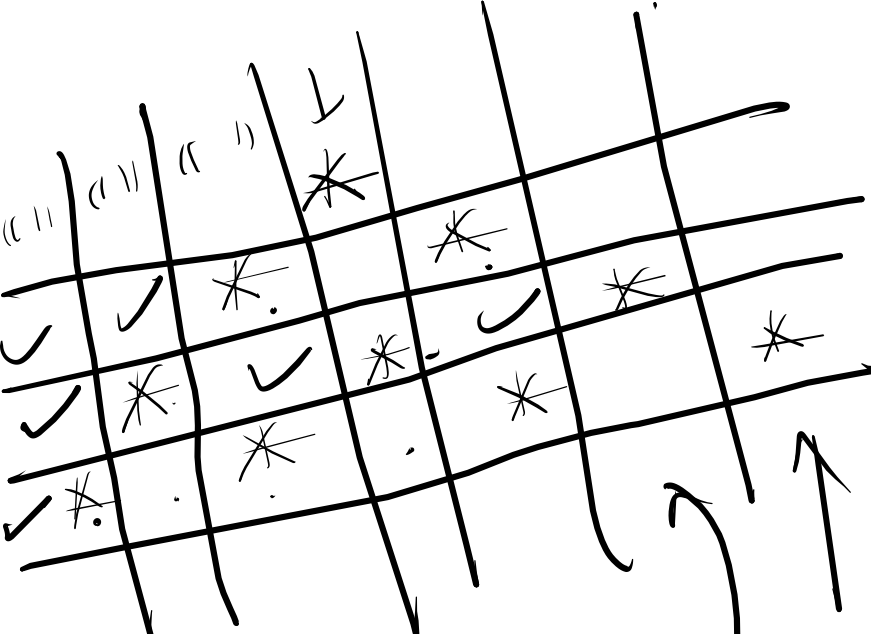
1
2
3

1 2 3

2
2 1

1 ... i

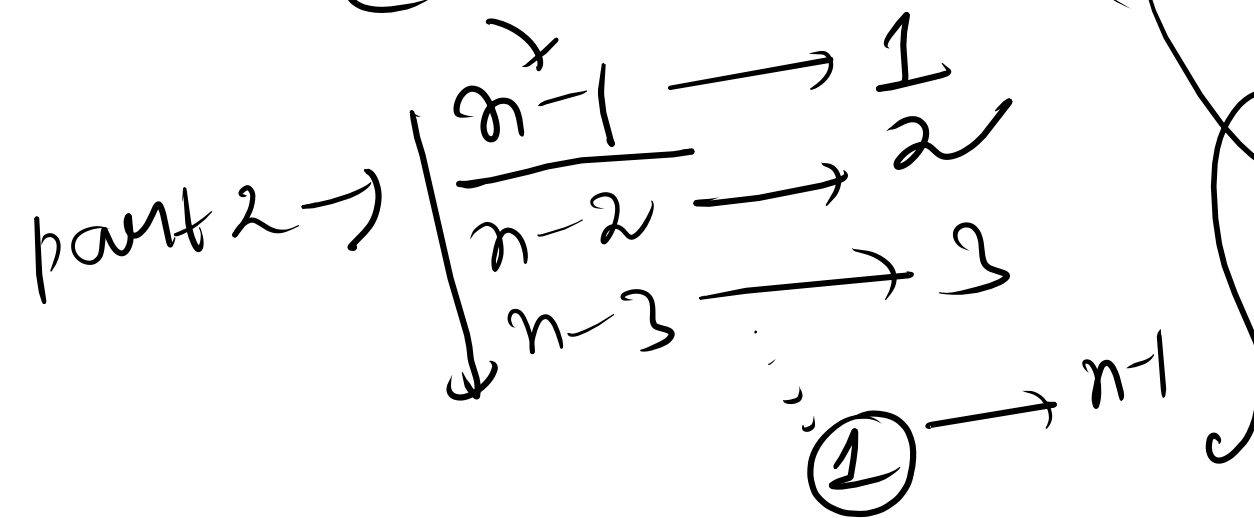
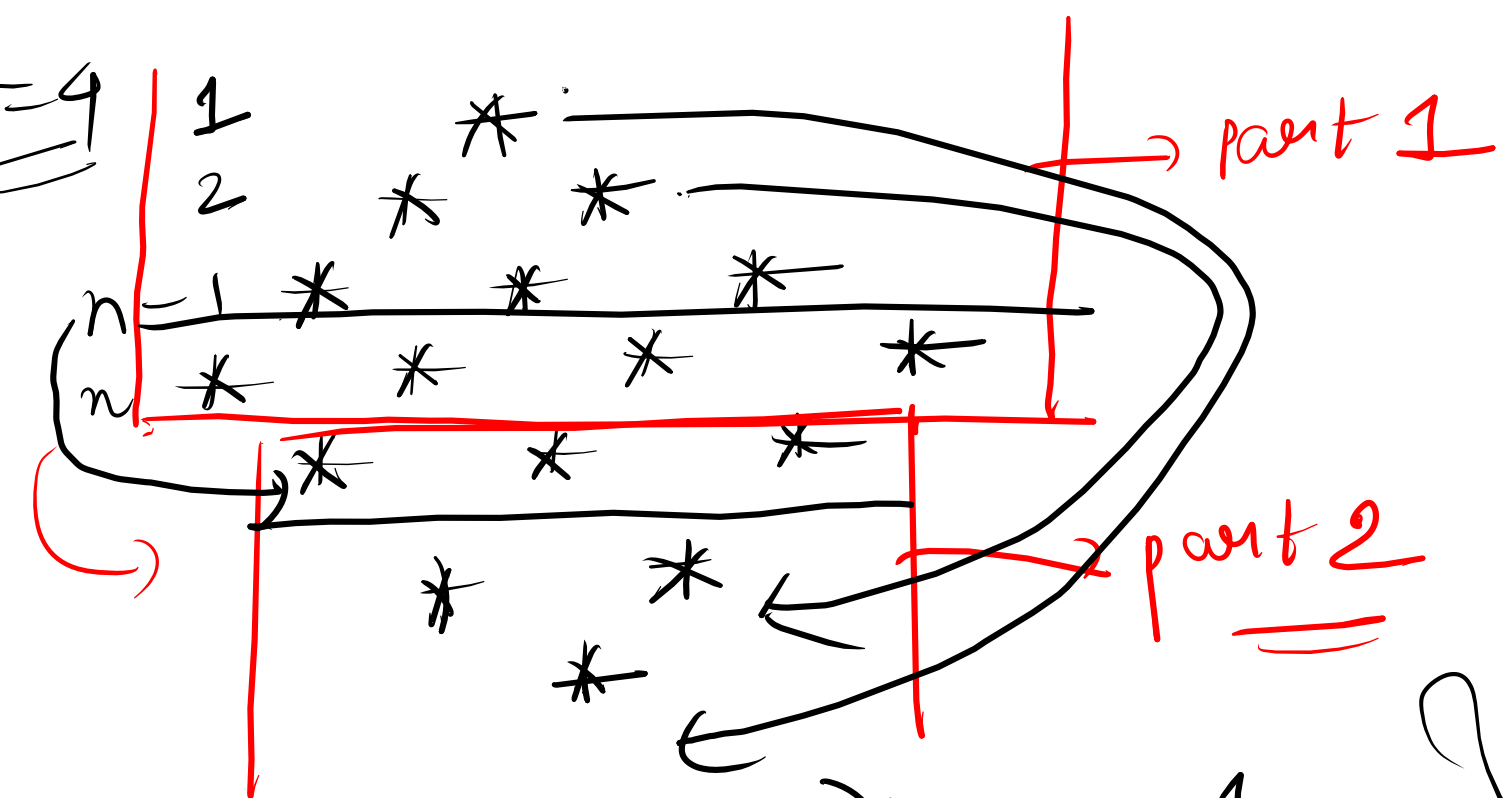
i-1 ... 1



$i = n - i$ spaces
 i stars



$n=4$



$n \times n$
chess board

$n=4$

W	B	W	B
B	W	B	W
W	B	W	B
B	W	B	W

$n=3$

	1	2	3
1	W	B	W
2	B	W	B
3	W	B	W

$(r+c) \% 2 == 0$
 even
 ↓ odd
 B

5 * 4

num = 1

~~n =~~ { n = 5 → rows
m = 4
↓
cols. }

① 2 3 4
5 6 7 8
9 0 1 2
3 4 5 6
7 8 9 0

1 2 3 4
5 6 7 8
9 0 1 2
3

5 * 4

num / 10

⑪ → 1
⑫ → 2
⑬ → 3

n x n

2.4 * 4

i/s

i = 1/n

i/s
i = 1/n
j = 1/n

	1			n
1	1	1	1	1
	1	0	0	1
	1	0	0	1
n	1	1	1	1

→ print

```
for(i=1; i<=n; i++) {
```

```
    for(j=1; j<=n; j++) {
```

```
        else {
```

```
            print('1');
```

```
        }  
        println();  
    }
```

```
        if(i==1 || i==n ||
```

```
           j==1 || j==n)
```

```
            print('0');
```